Topics Covered:

- Definitions
- Transformations

Additional Reading:

- Lynch, K.M. and Park, F.C. Modern Robotics: Section 2.1, 3.0-3.1
- Craig, J.J. Introduction to Robotics: Section 2.1-2.3
- Murray et al. A Mathematical Introduction to Robotic Manipulation: Section 2.1

Definitions

Definition: Rigid Body

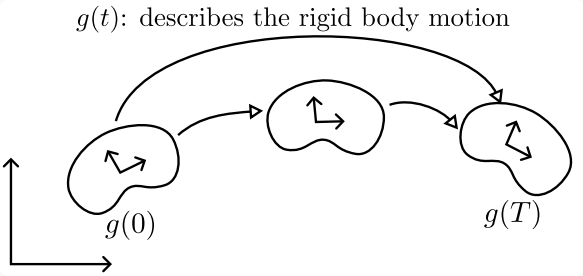
A rigid body is a collection of points/particles which have a fixed relationship amongst themselves.

- Concretely, the distance between any two particles never changes: $\|p_i - p_j\| = d_{ij}$ for all time, where each d_{ij} is a constant.
- This is what rules out bending, stretching, and squishing.

Definition: Rigid Body Motion

A rigid body motion describes how the individual particles of a rigid body move as a function of time. Equivalently, a rigid body motion is the motion of the body fixed reference frame.

- A body fixed frame is a coordinate frame that is glued to the body and moves with it. Because the body is rigid, every particle has the *same* coordinates in that frame for all time.
- $\implies$ tracking one frame is enough to track all (infinitely many) particles. This is why the rest of the course is about frames and not about particles.



Definition: Displacement/Transformation

A displacement or transformation is the movement or motion of a rigid body without reference to time-scale.

- from g(0) to g(T) is displacement
- Only the endpoints matter. Two very different motions (one direct, one that wanders around the room first) can produce the same displacement.

Definition: Planar Rigid Body Motion

A planar rigid body motion is rigid body motion during which all of the particles remain in a plane (or a set of parallel planes).
Stated precisely: a rigid body motion is planar if there exists a fixed plane P, with unit normal $\hat{n}$, such that every particle of the body stays a constant distance from P for all time.
Writing the motion as a rotation R together with a translation $\vec{d}$, this happens exactly when

1. $R\hat{n} = \hat{n}$ (the rotation is about an axis perpendicular to P), and
2. $\vec{d} \cdot \hat{n} = 0$ (the translation has no component out of P).

- Sketch proof: consider a particle at p sits a distance $p \cdot \hat{n}$ from P. The motion carries it to $Rp + \vec{d}$, which sits a distance

$$(Rp + \vec{d}) \cdot \hat{n} = p \cdot (R^\top \hat{n}) + \vec{d} \cdot \hat{n} = p \cdot \hat{n}$$

  from P, using $R^\top \hat{n} = \hat{n}$ from condition 1 and $\vec{d} \cdot \hat{n} = 0$ from condition 2. The distance never changes, so each particle is trapped in its own plane parallel to P.

- $\Longrightarrow$ this is where "a plane, or a set of parallel planes" comes from. A wheel rolling along a table is planar motion: each particle stays in its own vertical plane, and all of those planes are parallel.

Definition: Degrees of Freedom

A configuration of a rigid body is a specification of the positions of all the points of the body. The configuration space (C-space) $\mathcal{C}$ is the space of all configurations, and the degrees of freedom (DOF) is $\dim \mathcal{C}$, the minimum number of real numbers needed to represent a configuration.

$$\text{DOF} = \sum(\text{freedoms of the points}) \; - \; \#(\text{independent constraints}),$$

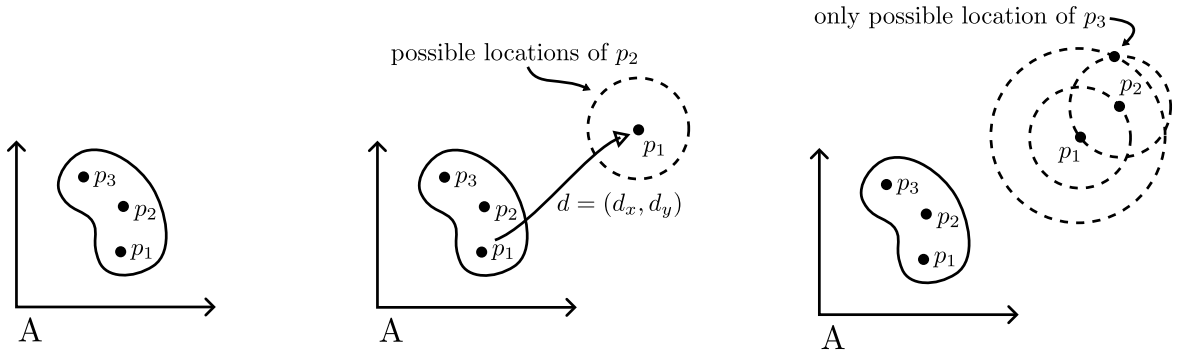
Example: a point on a circle, $\mathcal{C} = S^1$, $\dim \mathcal{C} = 1$.

coordinates	numbers used	constraints
ϕ	1	—
(x, y)	2	$x^2 + y^2 = 1$

Question: How many degrees of freedom in planar motion?

Proposition 1. *Planar motion has 3 degrees of freedom.*

Proof. Consider the planar rigid body illustrated in Figure 1.



(a) rigid body (b) constraints on p_1 and p_2 (c) constraints on p_3

Figure 1: Planar Rigid Body considered in Proposition 1.

1. The location of particle p_i denoted by (x_i, y_i). Lets consider how each reference point would be defined by a displacement of the rigid body as shown in the next figure.

2. Consider p_1. It has 2 degrees of freedom since $d = (d_x, d_y)$ is arbitrary.

3. consider p_2. It must satisfy some fixed relationship with respect to p_1, by definition of rigid body.
   $\implies$ i.e., fixed distance $(x_2 - x_1)^2 + (y_2 - y_1)^2 = d_{12}^2$.
   We started with 2 DOF for p_2 but we added one constraint
   $\implies$ 2 DOF - 1 constraint = 1 DOF.

4. Consider p_3. It must satisfy some fixed relationship with respect to p_1 and p_2.
   $\implies$ i.e., fixed distance $(x_3 - x_1)^2 + (y_3 - y_1)^2 = d_{13}^2$ and $(x_3 - x_2)^2 + (y_3 - y_2)^2 = d_{23}^2$.
   2 DOF - 2 constraints = 0 DOF.

5. consider $p_i, i > 3$.
   2 DOF, $i - 1$ constraints.
   $\implies$ 2 real constraints, $i - 3$ redundant ones
   $\implies$ 2 DOF - 2 constraints = 0 DOF.
   Why are the rest redundant? Fixing the distances to p_1 and p_2 already leaves only two possible locations for p_i, one on each side of the line through p_1 and p_2. The distance to p_3 picks which side, but picking a side is not a degree of freedom. Every distance after that is then automatically satisfied.

$\implies$ 3 total degrees of freedom. In general, will have $2n - (2(n-2)+1)$ constraints = 3 DOF. (for n particles).
Counting the constraints: p_1 has none, p_2 has 1, and each of the remaining $n - 2$ particles has 2, giving $1 + 2(n-2)$ in total.

■

> Meaning: If I want to describe the configuration associated to a planar rigid body, what do I need? $\implies$ (x, y, θ).

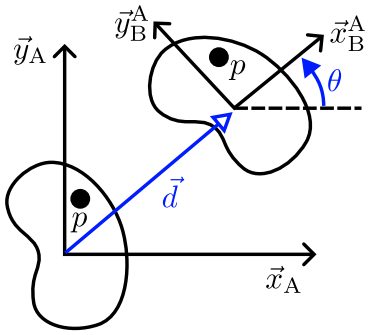
Coordinates

The 3DOF required to represent a planar rigid body are position and orientation. In a sense, coordinates also describe a displacement or transformation.

In the figure below there are two frames. Frame A is the fixed frame (also called the world or spatial frame), and frame B is the body frame that is glued to the body. The configuration g says where B is with respect to A:

- $\vec{d}$ is the vector from the origin of A to the origin of B, written in A coordinates. This is the position.
- θ is the angle from $\vec{x}_{\mathrm{A}}$ to $\vec{x}_{\mathrm{B}}$, measured counterclockwise. This is the orientation.

The same three numbers can be read two ways: as *where the body is* (a configuration), or as *how to get from frame A to frame B* (a displacement). We will use both readings constantly, so it is worth getting comfortable switching between them now.



Configuration of a rigid body:

$$g = (x, y, \theta)^{\top} \qquad \text{(Vector Notation)}$$

or equivalently:

$$g = (x, y, R(\theta))$$
$$g = (\vec{d}, R(\theta))$$
$$g = (\vec{d}, \theta)$$

Note: $R(\theta) := \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$

Note that we will use the notation $\vec{a}$ and curly brackets to denote vectors $\left(\vec{a} = \begin{Bmatrix} x \\ y \end{Bmatrix}\right)$.

$R(\theta)$ is our first example of the warning in the definition of degrees of freedom. It holds 4 numbers, but orientation in the plane is only 1 DOF. The 4 entries are not free: they are all built from the single angle θ. We will spend Lecture 2 working out exactly which constraints those 4 numbers satisfy, and why we would ever prefer 4 constrained numbers over 1 free one.